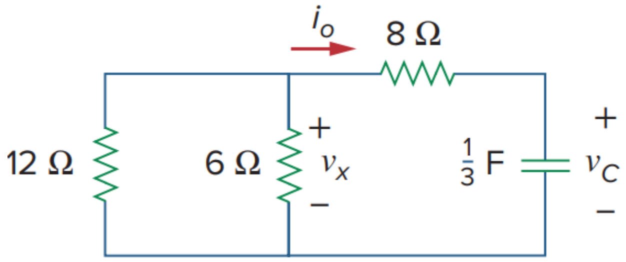
Solution to Problem 1



In this problem, we did not have to calculate values for t<0 as $V_{initial} = V_c(0)$ was given in the question

Given, $V_C(0) = 60\ V$

The equivalent resistance as seen from the capacitor terminal is,

$$R_{eq} = 8 + (6\,||\,12)$$

$$\gg R_{eq} = 12\ \Omega$$

Time constant,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 12 \times \frac{1}{3}$$

$$\gg \tau = 4\ s$$

As there is no source in the circuit,

$$V(\infty) = 0$$

So,

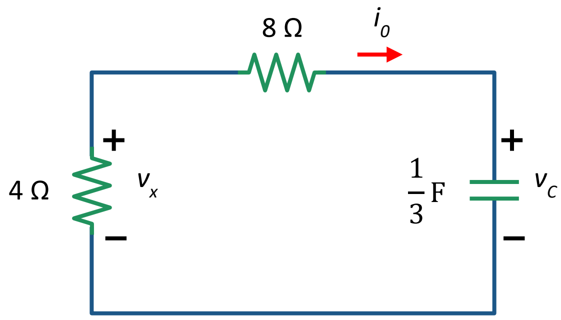
$$v_c(t) = V_c(0)e^{-\frac{t}{\tau}}$$

$$\gg v_c(t) = 60e^{-\frac{t}{4}}$$

$$\boxed{\gg v_c(t) = 60e^{-0.25t}\ (\mathrm{V})}$$

To find v_x and i_0 we can parallel, 12 Ω and 6 Ω making the circuit simplified as below.

$$12\,||\,6 = 4\ \Omega$$



Using voltage divider rule,

$$v_x = v_C(t) \times \frac{4}{4+8}$$

$$\gg v_x = 60e^{-0.25t} \times \frac{1}{3}$$

$$\boxed{\gg v_x = 20e^{-0.25t}\ (\text{V})}$$

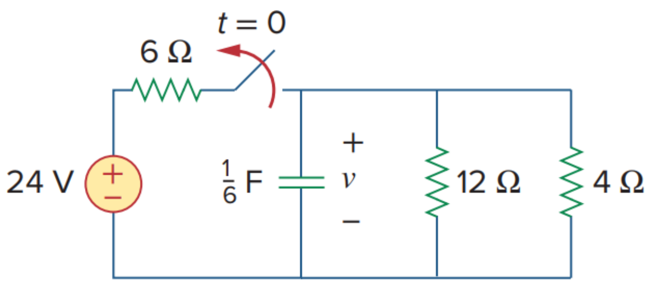
Applying KVL in the circuit,

$$4i_0 + 8i_0 + v_c = 0$$

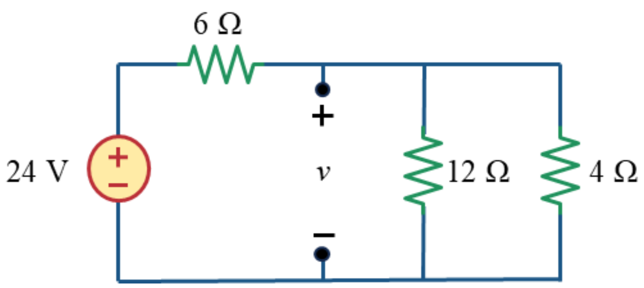
$$\gg i_0 = -\frac{1}{12} \times 60e^{-0.25t}$$

$$\boxed{\gg i_0 = -5e^{-0.25t}\ (\text{A})}$$

Solution to Problem 2



For $t < 0$, the switch is closed. With the capacitor open at dc, the circuit transforms into the one shown below

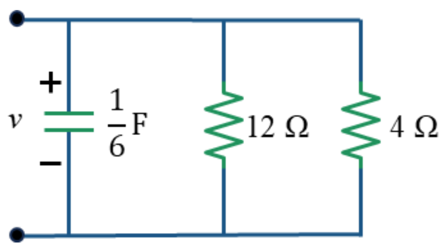


Now, using voltage divider rule, we can find the the initial voltage of the capacitor,

$$v(0) = 24 \times \frac{12 \,||\, 4}{(12 \,||\, 4) + 6}$$

$$v(0) = 8\ (\text{V})$$

For $t > 0$, the switch is open. The circuit transforms into the one shown below. As there is no independent source in the circuit, $V(\infty) = 0$



The Thevenin resistance as seen from the capacitor terminal,

$$R_{eq} = 12 \,||\, 4$$

$$\gg R_{eq} = 3\ \Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 3 \times \frac{1}{6}$$

$$\gg \tau = \frac{1}{2}\ s$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = v(0)e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 8e^{\frac{-t}{\frac{1}{2}}}$$

$$\boxed{\gg v(t) = 8e^{-2t}\ (\mathrm{V})}$$

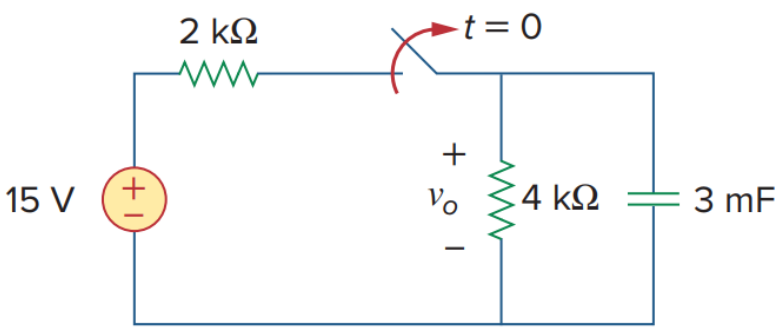
The initial energy stored in the capacitor is,

$$w_c(t) = \frac{1}{2}Cv(0)^2$$

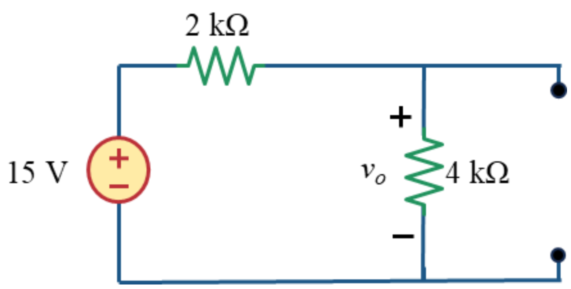
$$\gg w_c(t) = \frac{1}{2} \times \frac{1}{6} \times 8^2$$

$$\boxed{\gg w_c(t) = 5.33\ J}$$

Solution to Problem 3



For $t < 0$, the switch is closed. With the capacitor open at dc, the circuit transforms into the one shown below

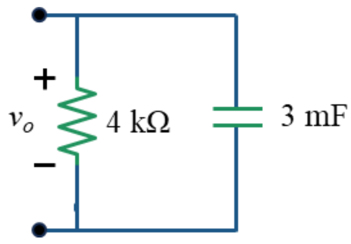


Using voltage divider rule, we can find the the initial voltage of the capacitor,

$$v_0(0) = 15 \times \frac{4}{2+4}$$

$$\gg v_0(0) = 10 \text{ (V)}$$

For $t > 0$, the switch is open. The circuit transforms into the one shown below. As there is no independent source in the circuit, $V(\infty) = 0$



The Thevenin resistance as seen from the capacitor terminal,

$$R_{eq} = 4\ k\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 4\ k\Omega \times 3\ mF$$

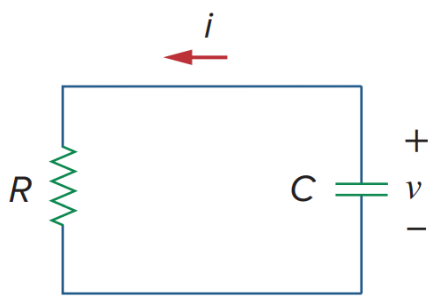
$$\gg \tau = 12\ s$$

So, the voltage across the capacitor for t > 0 is,

$$v_0(t) = v(0)e^{-\frac{t}{\tau}}$$

$$\boxed{\gg v_0(t) = 10e^{-\frac{t}{12}}}$$

Solution to Problem 4



Given in the question,

$$v(t) = 10e^{-4t}\ V$$

$$i(t) = 0.2e^{-4t} A$$

Applying KVL in the circuit,

$$i(t)R - v(t) = 0$$

$$\gg R = \frac{v(t)}{i(t)}$$

$$\gg R = \frac{10e^{-4t}}{0.2e^{-4t}}$$

$$\boxed{\gg R = 50\ \Omega}$$

From the equation of v, we can find the time constant,

$$\boxed{\tau = \frac{1}{4}\ s}$$

Since,

$$\tau = RC$$

$$\gg C = \frac{\tau}{R}$$

$$\gg C = \frac{\frac{1}{4}}{50}$$

$$\boxed{C = 5\ mF}$$

Since, the circuit does not have any source, Comparing the voltage with the general equation of capacitor voltage, which is,

$$v(t) = V_s + (V(0) - V_s)e^{-\frac{t}{\tau}}$$

Hence, initial voltage

$$V(0) = 10\ (\mathrm{V})$$

The initial energy stored in the capacitor is,

$$w_c(0) = \frac{1}{2}CV(0)^2$$

$$\gg w_c(0) = \frac{1}{2} \times (5 \times 10^{-3}) \times 10^2$$

$$\boxed{\gg w_c(0) = 0.25\ J}$$

To obtain the time it takes to dissipate 50% of the initial energy, we can set up the equation like below,

$$0.5w_c(0) = \frac{1}{2}C(V(t))^2$$

$$\gg 0.5\ \times 0.25 = \frac{1}{2} 5 \times 10^{-3} \times (10e^{-4t})^2$$

$$\gg 0.125 = 2.5 \times 10^{-3} \times 10^2 e^{-8t}$$

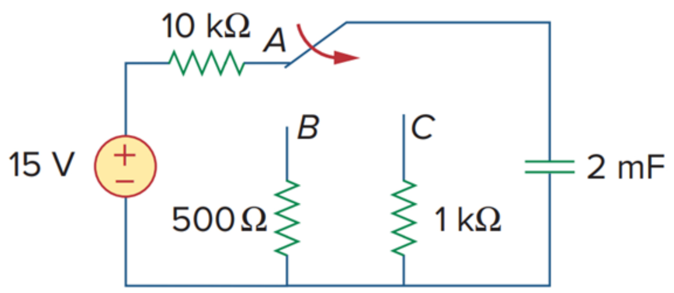
$$\gg e^{-8t} = \frac{1}{2}$$

$$\gg -8t = \ln\left(\frac{1}{2}\right)$$

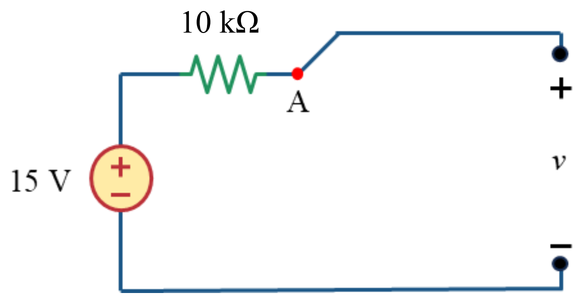
$$\boxed{\gg t = 86.64\ ms}$$

We used $0.5w_c(0)$ in place of $w(t)$ because we it says the capcitor energy has dissipated 50% of the initial energy. Meaning that at that specific time, $w_c(t)$ is $0.5w_c(0)$

Solultion to Problem 5



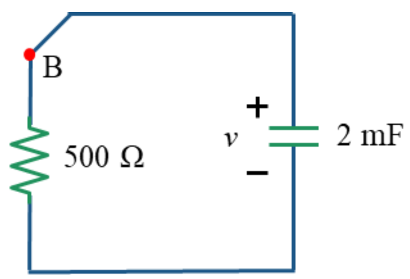
For $t < 0$, the switch is closed. With the capacitor open at dc, the circuit transforms into the one shown below,



As this is an open circuit, there will be no current flow through 10 kΩ resistor. Hence, the initial voltage of the capacitor,

$$v(0) = 15\ (\mathrm{V})$$

After the switch moved to point B, the circuit looks like below. And it remains like this for 1 s.



In this circuit, the Thevenin resistance as seen from the capacitor terminal,

$$R_{eq} = 500\ \Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 500\ \Omega \times 2\ mF$$

$$\gg \tau = 1\ s$$

As there are no independent sources in the circuit, After 1s, the voltage of the capacitor becomes,

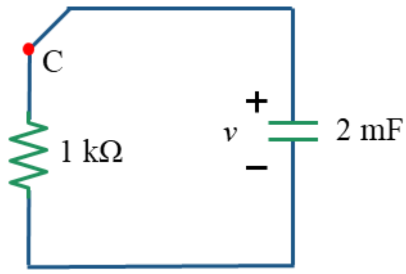
$$v(1) = v(0)e^{-\frac{1}{\tau}}$$

$$\gg v(1) = 15e^{-\frac{1}{1}}$$

$$v(1) = 5.518\ \text{V}$$

You might wonder, At point B, why the final voltage of capacitor is not 0. Where we saw earlier that when there is no voltage source in the circuit, the final voltage of capacitor becomes zero. However, in this case, the connection remains for only 1s and 5τ is 5s. Capacitor needs atleast 5τ to reach final value. So at point B, the capacitor did not have enough time to reach its final value of 0

After 1 s, the switch is moved to point C and the circuit becomes,



In this circuit, the Thevenin resistance as seen from the capacitor terminal,

$$R_{eq} = 1\ k\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 1\ k\Omega \times 2\ mF$$

$$\gg \tau = 2\ s$$

The initial voltage of the capacitor is now the final voltage for point B connection. So, V(0) = 5.518 V and as there is no independent source in the circuit, $V(\infty) = 0$.

So, the voltage across the capacitor at for point C connection,

$$v(t) = v(0)e^{-\frac{t-1}{\tau}}$$

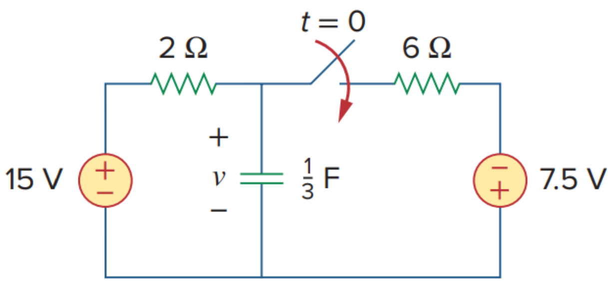
$$\gg v(t) = 5.518e^{-\frac{t-1}{2}}$$

$$\boxed{\gg v(t) = 5.518e^{-\frac{t-1}{2}}\ (\text{V})}$$

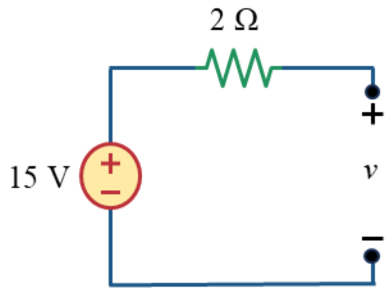
I know! Why $e^{-\frac{t-1}{\tau}}$ instead of $e^{-\frac{t}{\tau}}$ right? Well, it is because at the connection was made at point c at t = 1 s. Hence the -1.

Why -1 not +1? See the derivation of the formula.

Solultion to Problem 6



For $t < 0$, the switch is open. With the capacitor open at dc, the circuit transforms into the one shown.



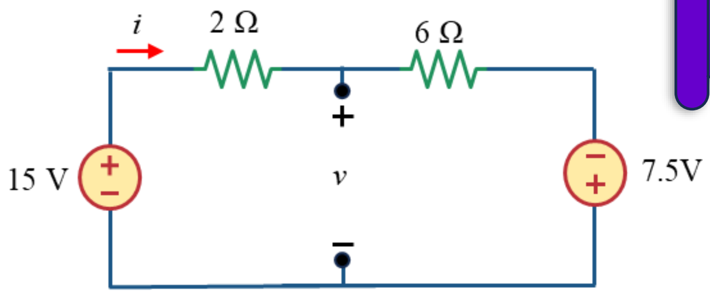
Here, the right portion of the circuit is not concerned because at t < 0, that portion is disconnected by the switch

There is an open circuit in the circuit. So no current flows through it and there is no voltage drop across 2 Ω. So,

$$V(0) = 15\ V$$

Why for final voltage of capacitor, we considered it open circuit? Because after t>0, eventually the capacitor becomes open circuit due to dc source connected

For $t > 0$, the switch is closed. The circuit transforms into the one shown below



As Voltage sources are connected in this circuit. We have to calculate the final voltage of capacitor.

Applying KVL in the circuit,

$$-15 + 2i + 6i - 7.5 = 0$$

$$\gg 8i = 22.5$$

$$\gg i = 2.8125\ A$$

You can also apply KVL in right portion of the circuit

To find the final capacitor voltage, applying KVL in the left portion of the circuit,

$$-15 + 2i + V(\infty) = 0$$

$$\gg -15 + 2 \times 2.8125 + V(\infty) = 0$$

$$\gg -15 + 5.625 + V(\infty) = 0$$

$$\gg V(\infty) = 9.375\ V$$

The Thevenin resistance as seen from the capacitor terminal (Omitting the independent sources: Replacing the voltage sources using short circuits and Current sources with open circuits)

$$R_{eq} = 2\ ||\ 6$$

$$R_{eq} = 1.5\ \Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 1.5\,\Omega \times \frac{1}{3}\,F$$

$$\gg \tau = 0.5\,s$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 9.375 + (15 - (9.375))e^{-\frac{t}{0.5}}$$

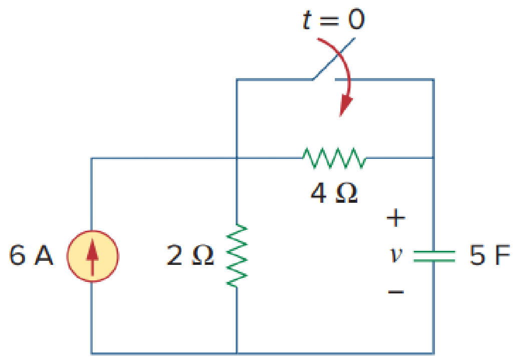
$$\boxed{\gg v(t) = 9.375 + 5.625e^{-2t}\ (\mathrm{V})}$$

Now, at t = 0.5s,

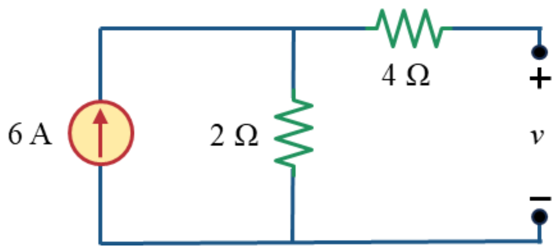
$$v(0.5) = 9.375 + 5.625e^{-2\times 0.5}$$

$$\boxed{v(0.5) = 11.444\,V}$$

Solultion to Problem 7



For $t < 0$, the switch is open. With the capacitor open at dc, the circuit transforms into the one shown



There is no current flow through the 4 Ω resistor. Hence,

$$V(0) = 6 \times 2$$

$$V(0) = 12\,V$$

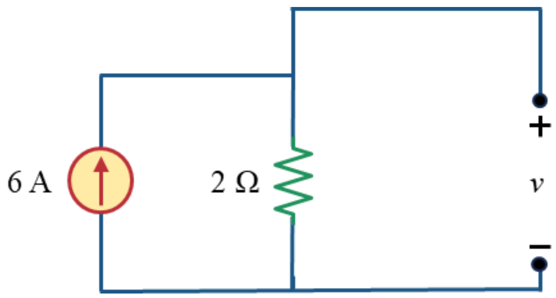
So, for t < 0, Capacitor voltage is,

$$\boxed{v(t) = 12\,V}$$

For $t > 0$, the switch is closed. The circuit transforms into the one shown below

Normally, for t < 0, we do not find out v(t). But the question demands it. Hence ...

Why for final voltage of capacitor, we considered it open circuit? Because after t > 0, eventually the capacitor becomes open circuit due to dc source connected



As current source is connected in this circuit. We have to calculate the final voltage of capacitor. We see that the capacitor is parallel to 2 Ω resistor. So calculating the voltage across 2 Ω will yield the final voltage of the capacitor.

$$V(\infty) = 6 \times 2$$

$$V(\infty) = 12 \text{ V}$$

The Thevenin resistance as seen from the capacitor terminal (Omitting the independent sources: Replacing the voltage sources using short circuits and Current sources with open circuits)

$$R_{eq} = 2\,\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 2\,\Omega \times 5\,F$$

$$\gg \tau = 10\,s$$

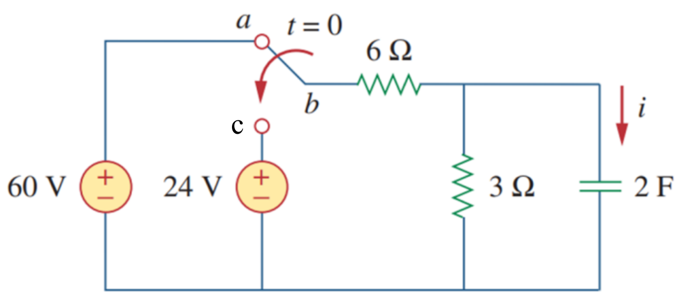
So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

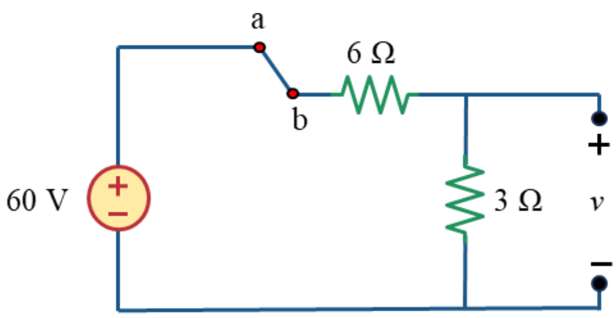
$$\gg v(t) = 12 + (12 - 12)e^{-\frac{t}{10}}$$

$$\boxed{\gg v(t) = 10\,V}$$

Solultion to Problem 8



For $t < 0$, the switch is connected to point a. With the capacitor open at dc, the circuit transforms into the one shown,

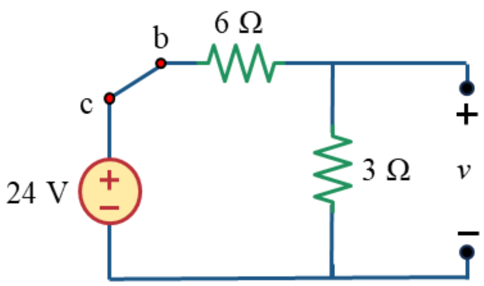


Using voltage divider rule,

$$V(0) = 60 \times \frac{3}{3+6}$$

$$V(0) = 20\ V$$

For $t > 0$, the switch is connected to point c. The circuit transforms into the one shown below



As voltage source is connected in this circuit. We have to calculate the final voltage of capacitor. We see that the capacitor is parallel to 3 Ω resistor. So calculating the voltage across 3 Ω will yield the final voltage of the capacitor.

Using Voltage divider rule,

$$V(\infty) = 24 \times \frac{3}{3+6}$$

$$V(\infty) = 8\ \mathrm{V}$$

The Thevenin resistance as seen from the capacitor terminal (Omitting the independent sources: Replacing the voltage sources using short circuits and Current sources with open circuits)

$$R_{eq} = 6\ ||\ 3$$

$$R_{eq} = 2\ \Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 2\ \Omega \times 2\ F$$

$$\gg \tau = 4\ s$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 8 + (20 - 8)e^{-\frac{t}{4}}$$

$$\gg v(t) = 8 + 12e^{-0.25t} \text{ (V)}$$

To find out i(t), Since,

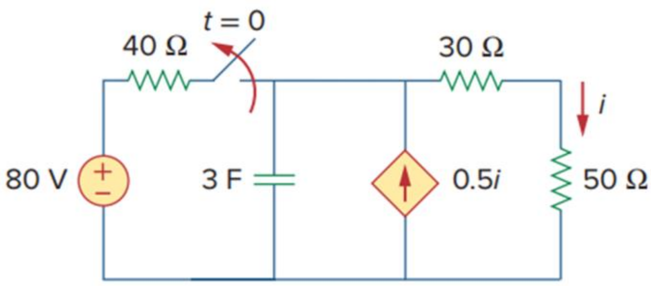
$$i(t) = C\frac{dv_c}{dt}$$

$$\gg i(t) = 2 \times \frac{d}{dt}(8 + 12e^{-0.25t})$$

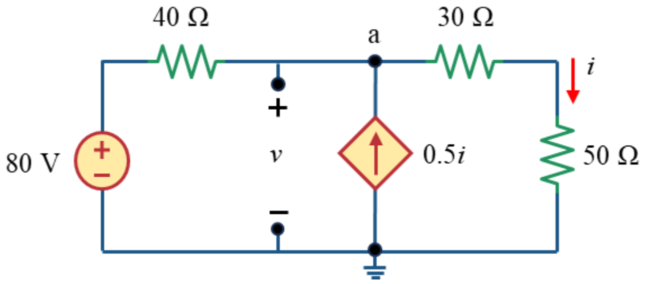
$$\gg i(t) = 2 \times (0 + (-0.25)12e^{-0.25t})$$

$$\boxed{\gg i(t) = -6e^{-0.25t}}$$

Solultion to Problem 9



For $t < 0$, the switch is closed. With the capacitor open at dc, the circuit transforms into the one shown.



From the circuit,

$$v_a = V(0)$$

Applying KCL at node a,

$$v_a\left(\frac{1}{40} + \frac{1}{80}\right) - \frac{80}{40} - 0.5i = 0$$

$$\gg v_a\left(\frac{1}{40} + \frac{1}{80}\right) - \frac{80}{40} - 0.5\frac{v_a}{80} = 0$$

$$\gg v_a\frac{3}{80} - 2 - \frac{v_a}{160} = 0$$

$$\gg \frac{v_a}{32} = 2$$

$$\gg v_a = 64\ V$$

$$V(0) = 64\ V$$

So, for t < 0, Capacitor voltage is,

$$v(t) = 64\ V$$

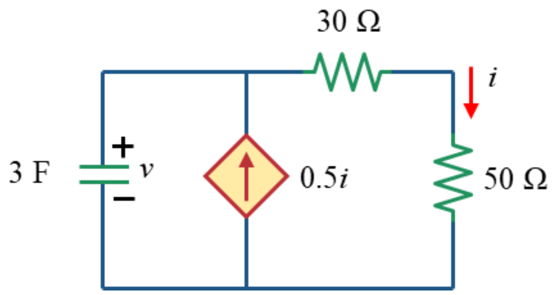
So for t < 0, the current is,

$$i = \frac{64}{80} A$$

$$\boxed{i = 0.8\ A}$$

Normally, for t < 0, we do not find out v(t). But the question demands it. Hence ...

For $t > 0$, the switch is open. The circuit transforms into the one shown below. As there is no independent source in the circuit, $V(\infty) = 0$



The Thevenin resistance as seen from the capacitor terminal (Omitting the independent sources: Replacing the voltage sources using short circuits and Current sources with open circuits). As there is a dependent source, we use a 1A current source inplace of the capacitor to find out the voltage across the current source. Then using R = V/I, we can find out the thevenin resistance.

Dependent source makes life harder for R_{th} 😤

Applying KCL at the top node,

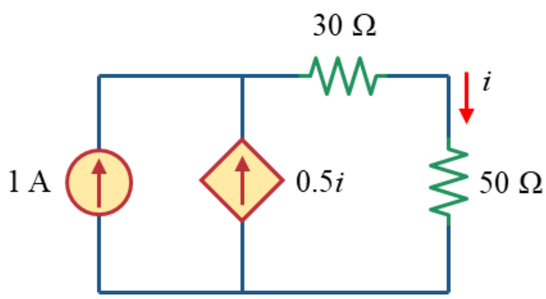
$$1 + 0.5i = i$$

$$i = 2\ A$$

Now, the voltage across the 1A current source, (Same as the voltage across the 80 Ω resistor)

$$V_{1A} = 2 \times 80$$

$$V_{1A} = 160\ V$$



Now, the thevenin resistor,

$$R_{th} = \frac{V_{1A}}{1}$$

$$R_{th} = 160\ \Omega$$

The time constant is,

$$\tau = R_{th} \times C$$

$$\gg \tau = 160\ \Omega \times 3\ F$$

$$\gg \tau = 480\ s$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(0)e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 64e^{-\frac{t}{480}}$$

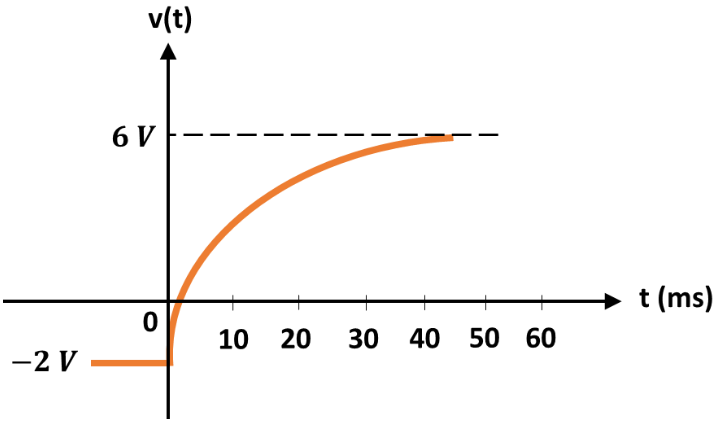
Finally, for t > 0, the current is,

$$i(t) = \frac{v(t)}{80}$$

$$\gg i(t) = \frac{64e^{-\frac{t}{480}}}{80}$$

$$\boxed{\gg i(t) = 0.8e^{-\frac{t}{480}}}$$

Solultion to Problem 10



The figure below shows the voltage response of an RC circuit to a sudden DC voltage applied through an equivalent resistance of 4 $k\Omega$

From the graph we can find the initial voltage of the capacitor at t = 0,

$$V(0) = -2\ V$$

We can also find the final voltage of the capacitor as,

$$V(\infty) = 6\ \text{V}$$

We can see that at about 45 s, the graph reaches steady state. Hence the time is 5τ.

$$5\tau = 45$$

$$\boxed{\gg \tau = 9\ ms}$$

So the value of the capacitor,

$$C = \frac{\tau}{R}$$

$$\gg C = \frac{9\ ms}{4k\Omega}$$

$$C = 2.25\ uF$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 6 + (-2 - 6)e^{-\frac{t}{9\times10^{-3}}}$$

$$\boxed{\gg v(t) = 6 - 8e^{-\frac{1000}{9}t}\ (\text{V})}$$

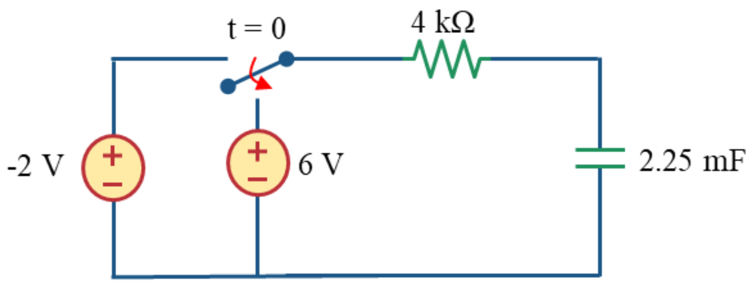
The initial energy stored in the capacitor is,

$$w_c(t) = \frac{1}{2}CV(0)^2$$

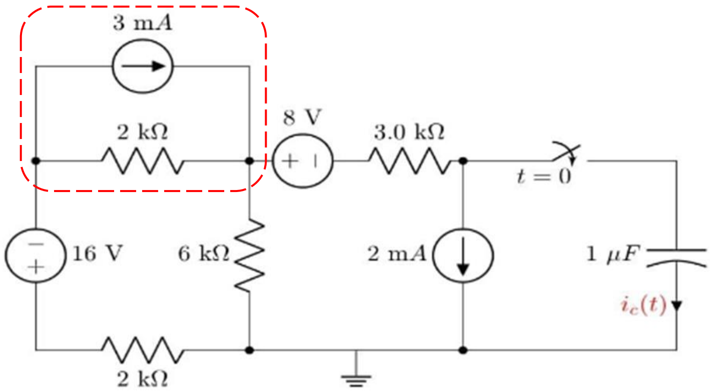
$$\gg w_c(t) = \frac{1}{2} \times 2.25 \times 10^{-6} \times (-2)^2$$

$$\boxed{\gg w_c(t) = 4.5 \times 10^{-6}\ J}$$

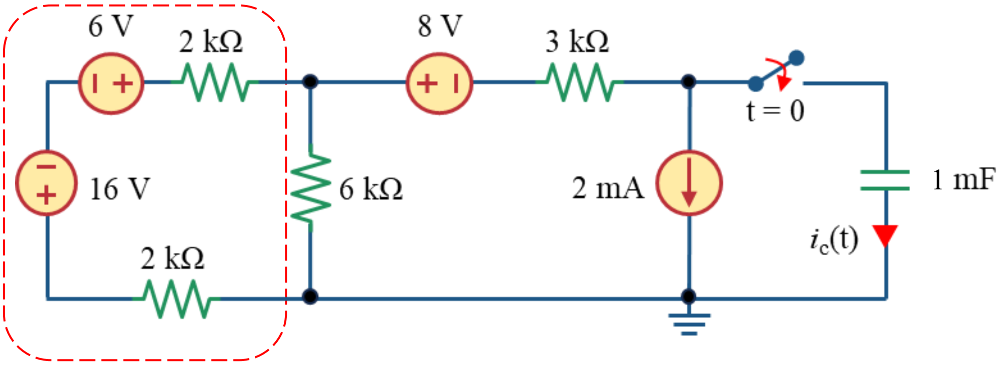
The circuit is,



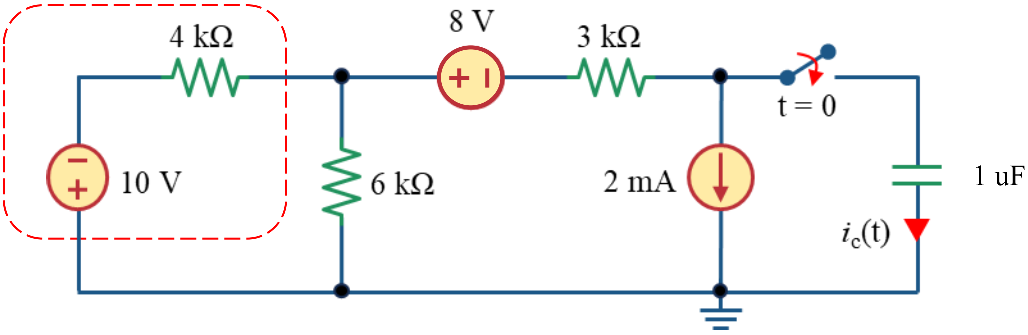
Solultion to Problem 11



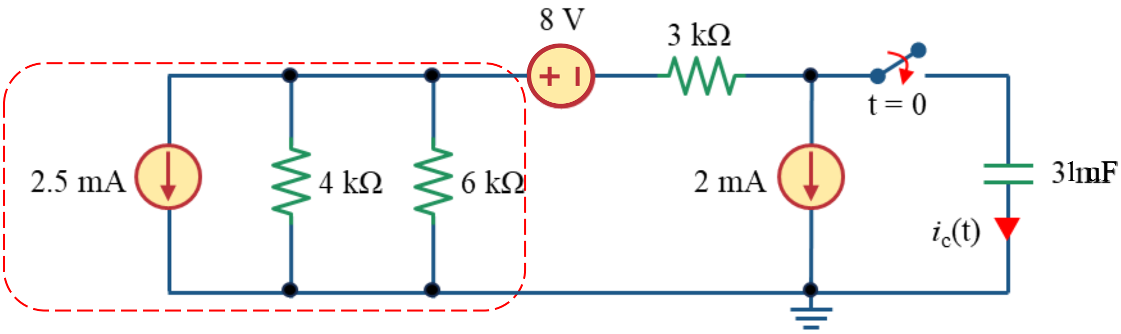
We have to use source transformation to simplify the circuit. First transforming the current source in marked section into voltage source.



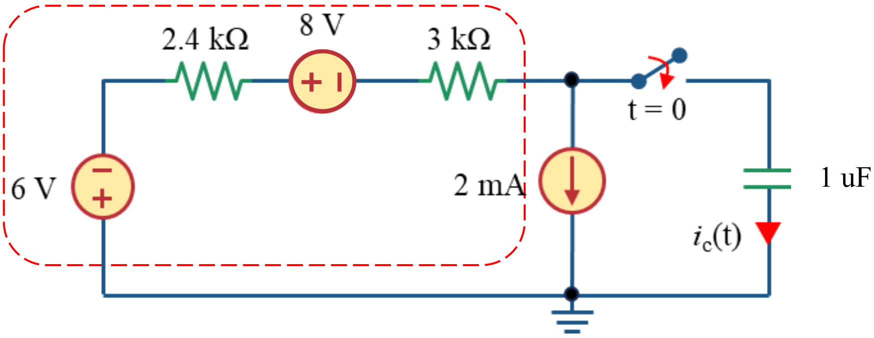
Then, let us find the equivalent voltage source and resistor for the marked section



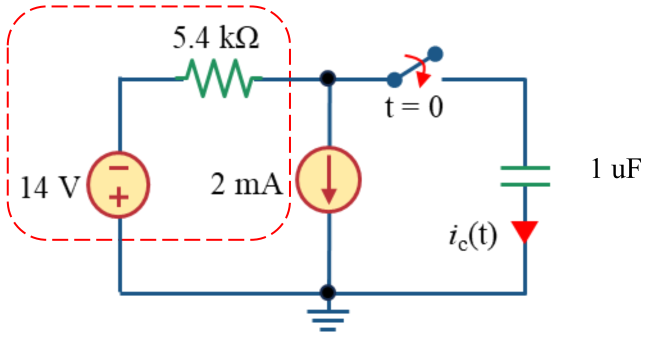
Transforming the voltage source in the marked section,



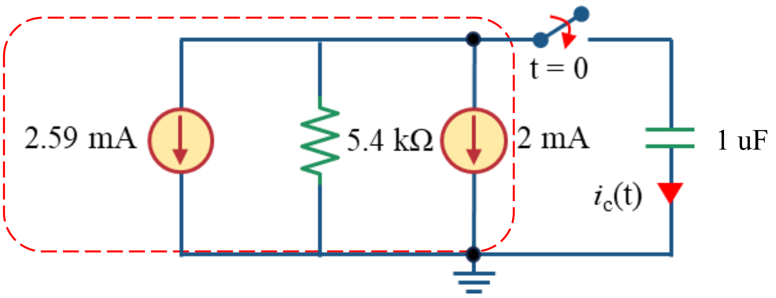
Calculating the parallel equivalent resistance of the circuit in marked section ($4 \parallel 6 = 2.4\ k\Omega$) and transforming the current source,



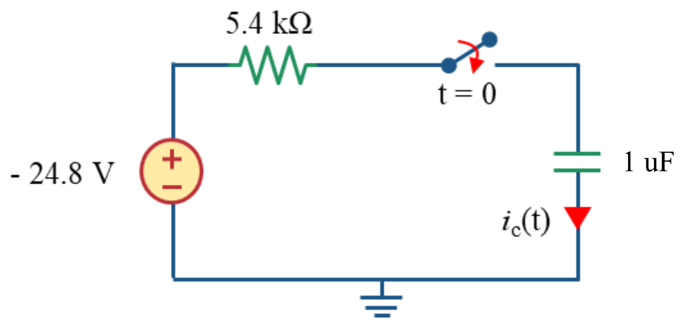
let us find the equivalent voltage source and resistor for the marked section



Transforming the voltage source in the marked section,



Finding the equivalent current source and transforming it,



Comparing with the second circuit of the question,

$$V_1 = -24.8\,V$$

$$R_1 = 5.4\,k\Omega$$

As the switch closes at t = 0, looking at the circuit, the capacitor was not connected to any voltage before t = 0. So the capacitor does not have any initial voltage.

$$V(0) = 0\,V$$

The Thevenin resistance as seen from the capacitor terminal

$$R_{eq} = 5.4\,k\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 5.4\,k\Omega \times 1\,uF$$

$$\gg \tau = \frac{5.4}{1000}\ s$$

The final voltage of the capacitor will be the voltage of the voltage source

$$V(\infty) = -24.8\ V$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = -24.8 + (0 - (-24.8))e^{-\frac{t}{\frac{5.4}{1000}}}$$

$$\gg v(t) = -24.8 + 24.8e^{-\frac{1000}{5.4}t}\ \text{(V)}$$

Applying KVL in the circuit,

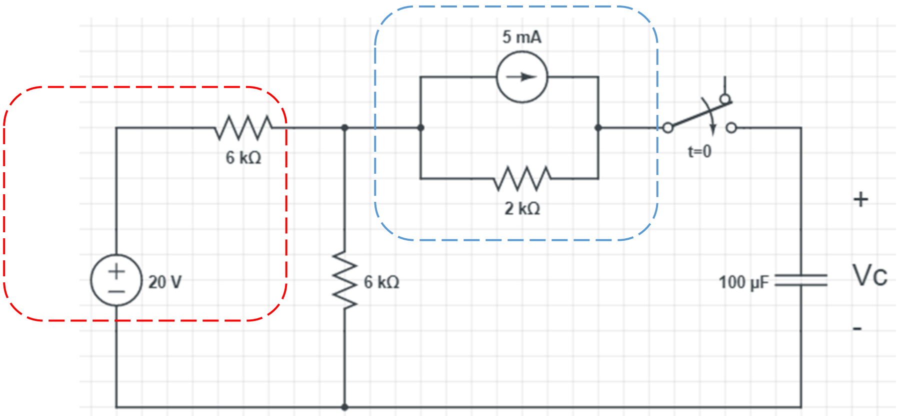
$$-(-24.8) + 5i(t) + v(t) = 0$$

$$\gg i(t) = -\frac{v(t) + 24.8}{5}$$

$$\gg\ i(t) = -\frac{-24.8 + 24.8e^{-\frac{1000}{5.4}t} + 24.8}{5}$$

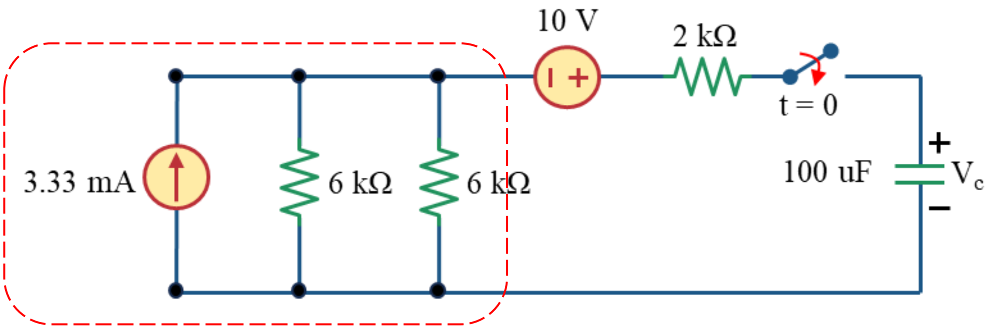
$$\boxed{i(t) = -4.96e^{-\frac{1000}{5.4}t}}$$

Solultion to Problem 12

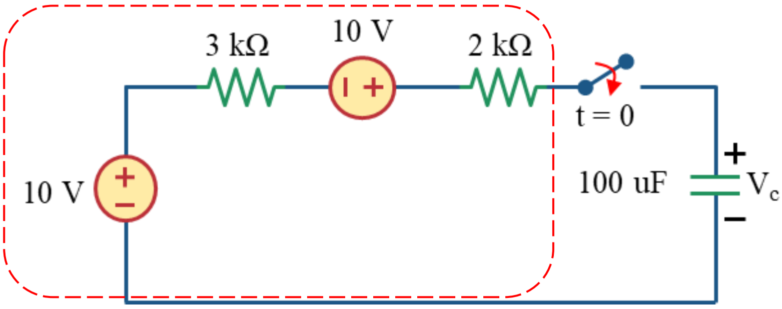


We have to use source transformation to simplify the circuit.

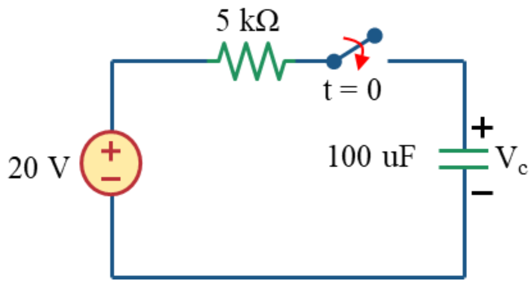
First transforming the voltage source in red marked section into current source as well as the current source in blue marked section into voltage source.



Calculating the parallel equivalent resistance of the circuit in marked section ($6 \parallel 6 = 3\ k\Omega$) and transforming the current source,



Finding the equivalent voltage source and resistance in the marked section will get us the simplifed circuit



Comparing with the second circuit of the question,

$$V_1 = 20\ V$$

$$R_1 = 5\ k\Omega$$

As the switch closes at t = 0, looking at the circuit, the capacitor was not connected to any voltage before t = 0. So, the capacitor does not have any initial voltage.

$$V(0) = 0\ V$$

The Thevenin resistance as seen from the capacitor terminal

$$R_{eq} = 5\ k\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 5\ k\Omega \times 100\ uF$$

$$\gg \tau = 0.5\ s$$

The final voltage of the capacitor will be the voltage of the voltage source

$$V(\infty) = 20\ V$$

So, the voltage across the capacitor for t > 0 is,

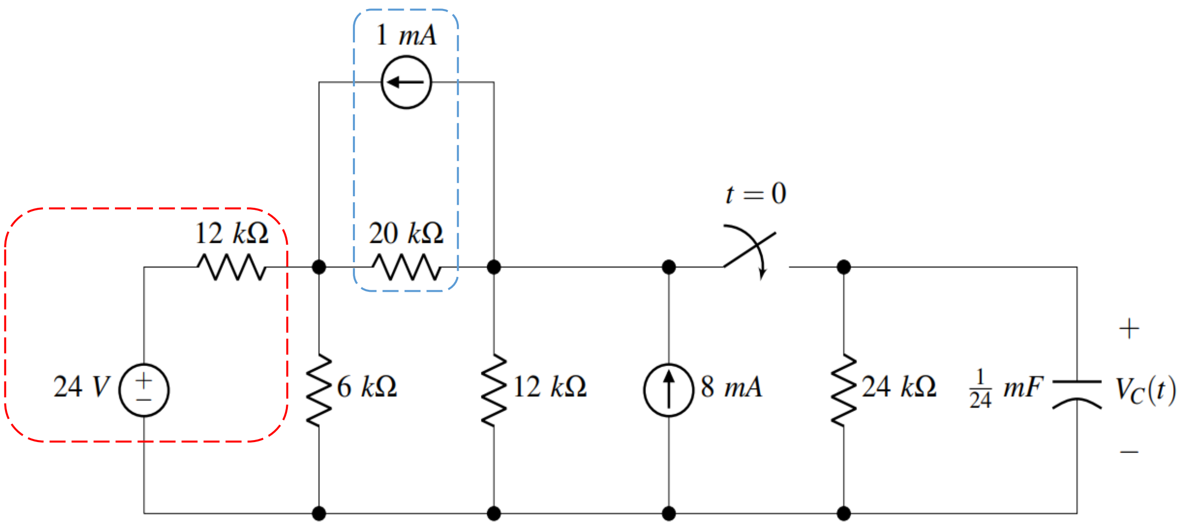
$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 20 + (0 - 20)e^{-\frac{t}{0.5}}$$

$$\gg v(t) = 20 - 20e^{-2t}\ (\text{V})$$

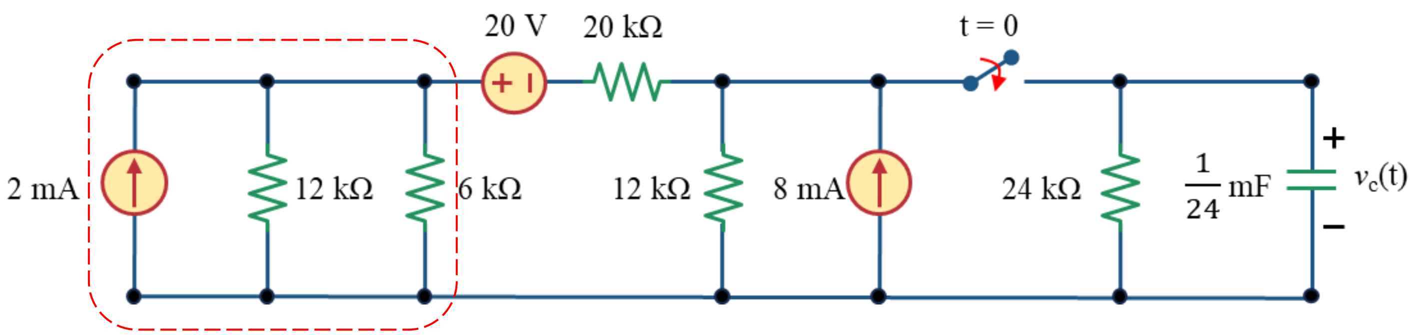
$$\boxed{\gg v(t) = 20(1 - e^{-2t})\ (\text{V})}$$

Solultion to Problem 13

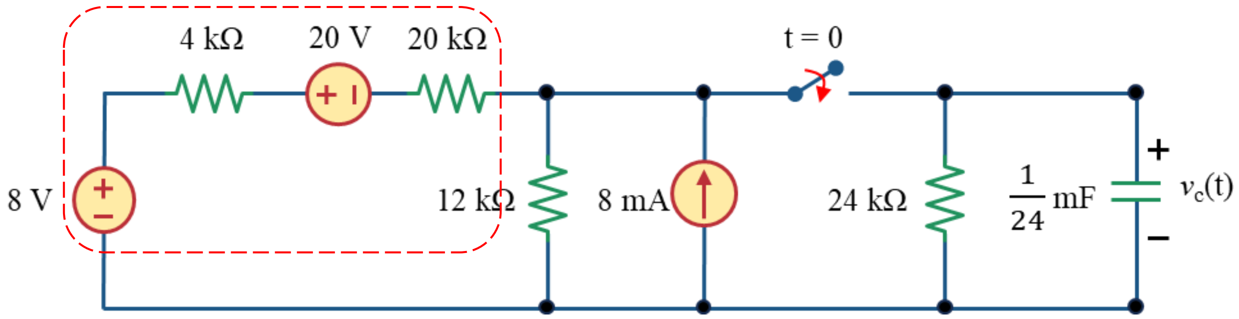


We have to use source transformation to simplify the circuit.

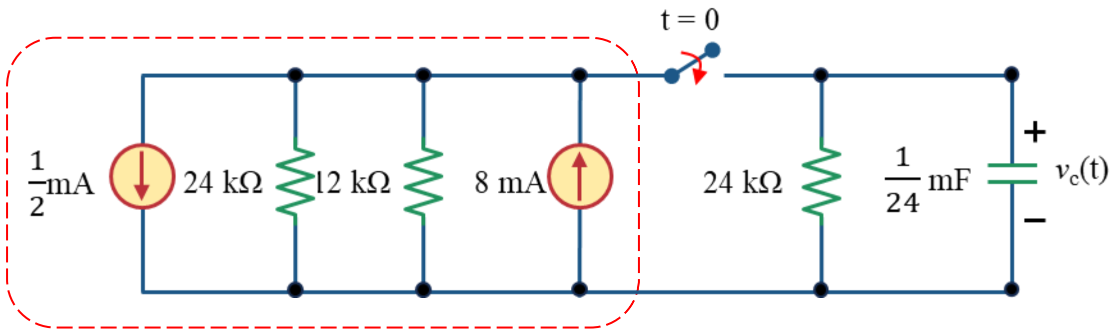
First transforming the voltage source in red marked section into current source as well as the current source in blue marked section into voltage source.



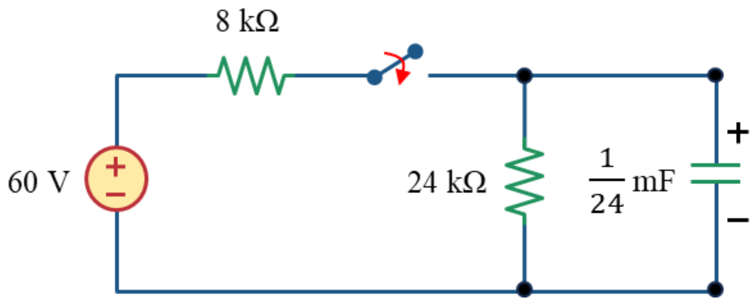
Calculating the parallel equivalent resistance of the circuit in marked section (12 || 6 = 4 $k\Omega$) and transforming the current source,



Finding the equivalent voltage source and resistance in the marked section and transforming the equivalent voltage source into current source



Calculating the parallel equivalent resistance of in marked section (24 || 12 = 8 $k\Omega$) and equivalent current source and transforming the equivalent current source into voltage source. This will get us the simplified circuit.



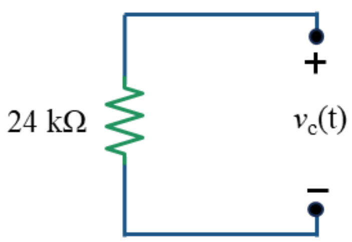
Comparing with the second circuit of the question,

$$V_1 = 60\ V$$

$$\boxed{R_1 = 8\ k\Omega}$$

Now, let us find out the capacitor voltage

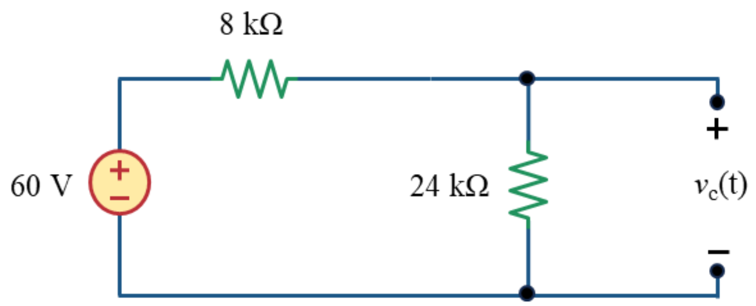
For $t < 0$, the switch is open. With the capacitor open at dc, the circuit transforms into the one shown.



As the switch closes at t = 0, looking at the circuit, the capacitor was not connected to any voltage before t = 0. So, the capacitor does not have any initial voltage.

$$V(0) = 0\,V$$

For $t > 0$, the switch is closed. The circuit transforms into the one shown below



As Voltage sources are connected in this circuit. We have to calculate the final voltage of capacitor.

Using Voltage divider rule,

$$V(\infty) = 60 \times \frac{24}{24+8}$$

$$V(\infty) = 45\text{ V}$$

The Thevenin resistance as seen from the capacitor terminal (Omitting the independent sources: Replacing the voltage sources using short circuits and Current sources with open circuits)

$$R_{eq} = 8\,||\,24$$

$$R_{eq} = 6\,k\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 6\,k\Omega \times \frac{1}{24}\,F$$

$$\gg \tau = \frac{1}{4}\,s$$

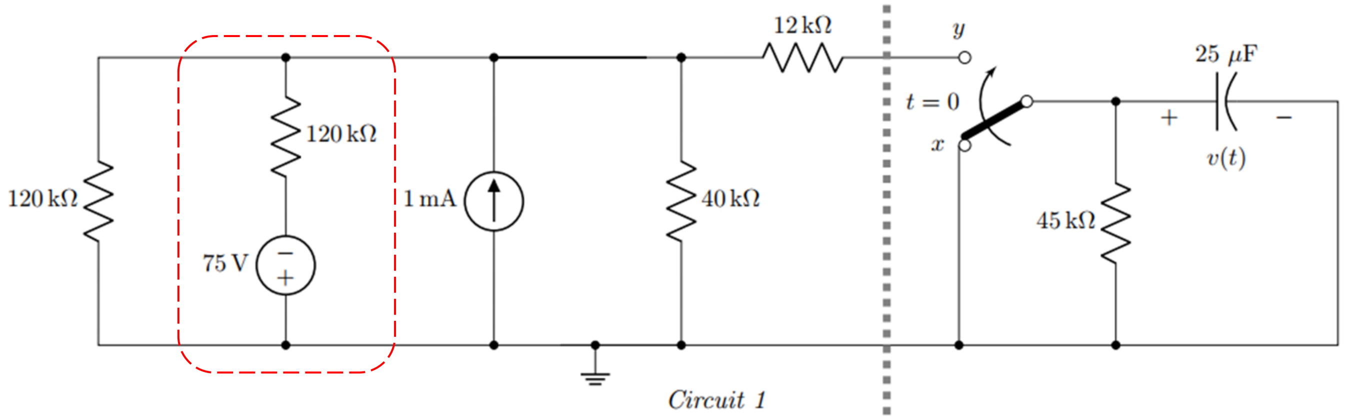
So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 45 + (0 - 45)e^{-4t}$$

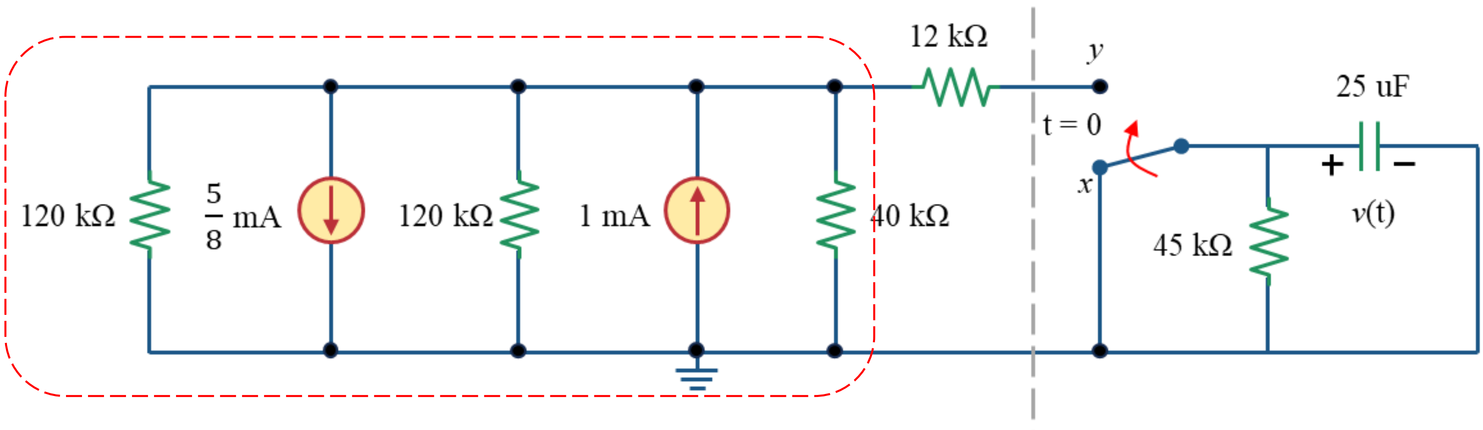
$$\boxed{\gg v(t) = 45\,(1 - e^{-4t})\,(V)}$$

Solultion to Problem 14

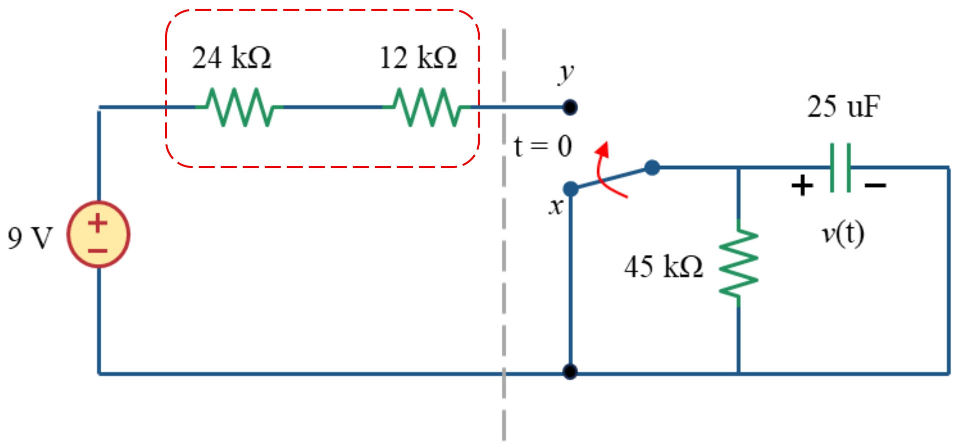


We have to use source transformation to simplify the circuit.

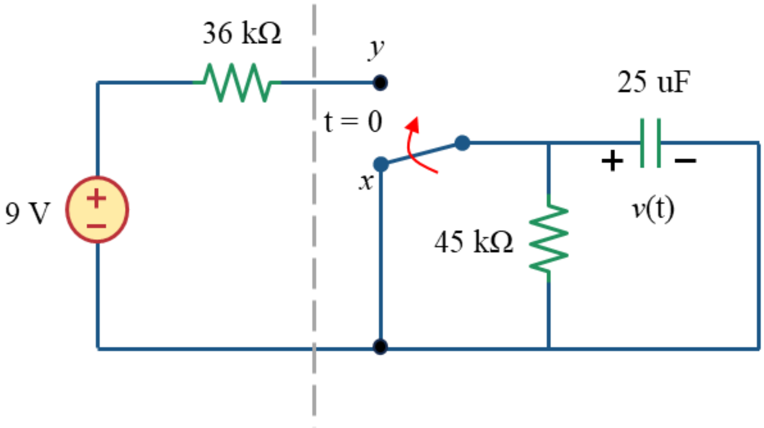
First transforming the voltage source in red marked section into current source



Calculating the parallel equivalent resistance of in marked section ($120 \parallel 120 \parallel 40 = 24\ k\Omega$) and equivalent current source and transforming the equivalent current source into voltage source.

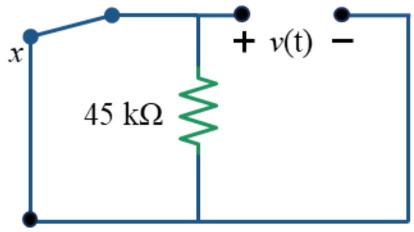


Replacing the red marked resistance with their equivalent resistance will yield the simplified circuit as shown below



Now, let us find out the capacitor voltage

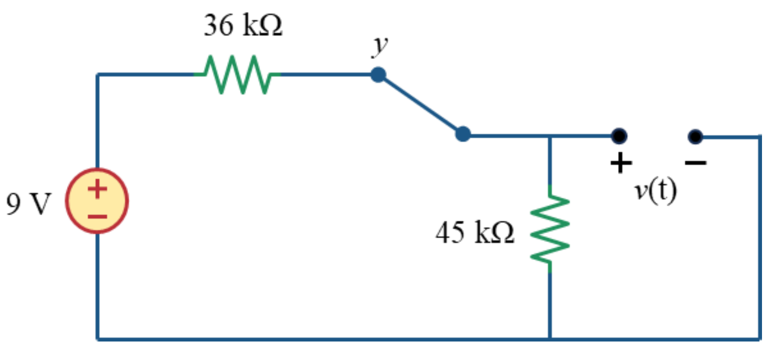
For $t < 0$, the switch is at point x. With the capacitor open at dc, the circuit transforms into the one shown.



As the switch closes at t = 0, looking at the circuit, the capacitor was not connected to any voltage before t = 0. So, the capacitor does not have any initial voltage.

$$V(0) = 0\,V$$

For $t > 0$, the switch is closed. The circuit transforms into the one shown below



As Voltage sources are connected in this circuit. We have to calculate the final voltage of capacitor.

Using Voltage divider rule,

$$V(\infty) = 9 \times \frac{45}{16 + 45}$$

$$V(\infty) = 5\text{ V}$$

The Thevenin resistance as seen from the capacitor terminal (Omitting the independent sources: Replacing the voltage sources using short circuits and Current sources with open circuits)

$$R_{eq} = 36 \,||\, 45$$

$$R_{eq} = 20\ k\Omega$$

The time constant is,

$$\tau = R_{eq} \times C$$

$$\gg \tau = 20\ k\Omega \times 25\ uF$$

$$\gg \tau = 0.5\ s$$

So, the voltage across the capacitor for t > 0 is,

$$v(t) = V(\infty) + (V(0) - V(\infty))e^{-\frac{t}{\tau}}$$

$$\gg v(t) = 5 + (0 - 5)e^{-\frac{t}{0.5}}$$

$$\boxed{\gg v(t) = 5\,(1 - e^{-2t})\ (V)}$$